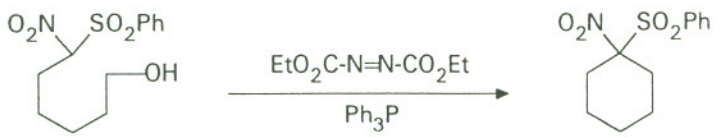
ORGANIC CUMULATIVE EXAMINATION
December 4, 1993

Propose a reasonable step-by-step mechanism for three of the following reactions. Show the structural formula for each important mechanistic intermediate. Describe how your mechanism explains any regioselectivity and/or stereoselectivity if that is pertinent. (Reagents and solvents shown over and under the same arrow are simultaneously present. Reagents shown over and under a second arrow are added later, but without isolation of a product from the reaction brought about by the first arrow. All reactions are worked up in the usual manner.) **Show clearly which three reactions you want graded. If you do not, reactions 1, 2, and 3 will be graded.**

1.



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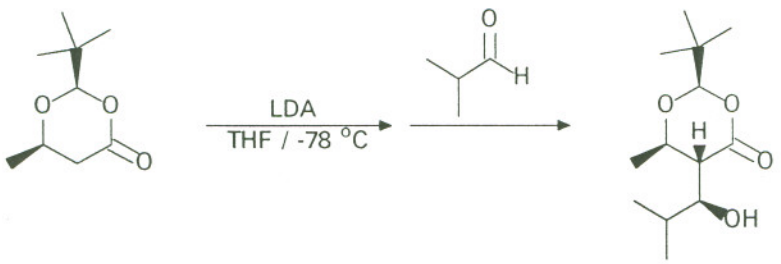

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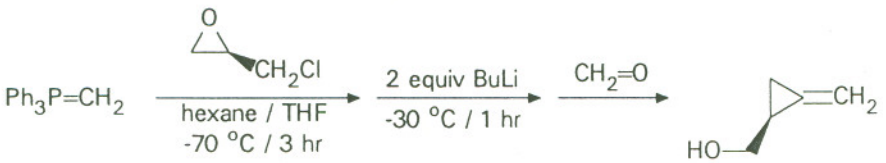

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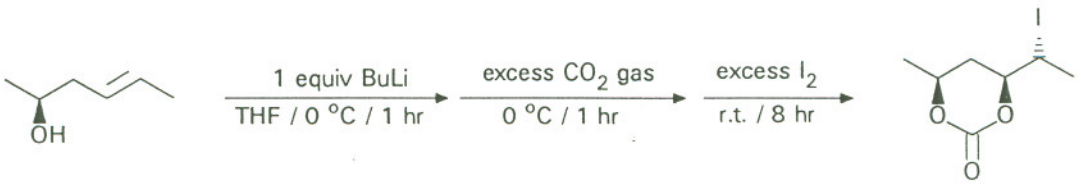

References to answers:

1. Yu, J.; Cho, H-S.; Falck, J. R. *J. Org. Chem.* **1993**, *58*, 5892-5894.
2. Seebach, D.; Lapierre, J-M.; Jaworek, W.; Seiler, P. *Helv. Chim. Acta* **1993**, *76*, 459-475.
3. Okuma, K.; Tanaka, Y.; Yoshihara,K.; Ezaki, A.; Koda, G.; Ohta, H.; Hara, K.; Kashimura, S. *J. Org. Chem.* **1993**, *58*, 5915-5917.
4. Tirado, R.; Prieto, J. A. *J. Org. Chem.* **1993**, *58*, 5666-5673.